
ACCOUNTING FOR STATE PREDICTION UNCERTAINTY IN DELAYED REINFORCEMENT LEARNING

A PREPRINT

tujunqi liyan

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ABSTRACT

1 Brief version

For state prediction, instead of outputting a single point, we consider the uncertainty of the prediction and output a Gaussian state. augmented state $\tilde{S}_t = (S_{t-\Delta T}, a^{(t-\Delta T):(t-1)})$ predicted state $\hat{S}_t = (\hat{S}_t, \sigma^2)$ sample N times S from distribution of $S^i = \text{sample}(N(\tilde{S}))$

$$\text{Var}[Q(\tilde{S}, a)] \approx \sigma^2 + \frac{1}{N} \sum_{t=1}^N \hat{Q}(s^i, a)^\top \hat{Q}(s^i, a) - \hat{Q}(\hat{s}, a)^\top \hat{Q}(\hat{s}, a)$$

new delay policy can be defined like:

$$\begin{aligned} \pi'(a|\tilde{s}) &= \frac{\beta}{\text{Var}[Q^{\pi'}(\tilde{s}, a)]} \pi(a|\hat{s})/Z(\tilde{s}); \\ Z(\tilde{s}) &= \int_a \text{Var}[Q^{\pi'}(\tilde{s}, a)] \pi(a|\hat{s}) da \end{aligned}$$

combined our new delayed policy with SAC algorithm, we can get like: critic use no delay data:

$$J_Q(\theta) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{D}} \left[(Q_\theta(s_t, a_t) - y(r_t, s_{t+1}))^2 \right] \quad (1)$$

$$y(r_t, s_{t+1}) = r_t + \gamma \mathbb{E}_{a_{t+1} \sim \pi'(\cdot|s_{t+1})} (Q_{\theta \text{ targ}}(S_{t+1}, a_{t+1}) - \alpha \log \pi'(a_{t+1}|s_{t+1})) \quad (2)$$

can be proved that s_{t+1} is not noisy: $\pi'(\cdot|s_{t+1}) = \pi(\cdot|s_{t+1})$ so:

$$y(r_t, s_{t+1}) = r_t + \gamma \mathbb{E}_{a_{t+1} \sim \pi(\cdot|s_{t+1})} (Q_{\theta \text{ targ}}(S_{t+1}, a_{t+1}) - \alpha \log \pi(a_{t+1}|s_{t+1})) \quad (3)$$

$$\nabla_\theta J_Q(\theta) = \nabla_\theta Q_\theta(a_t, s_t) (Q_\theta(s_t, a_t) - (r_t + \gamma \mathbb{E}_{a_{t+1} \sim \pi(\cdot|s_{t+1})} (Q_{\theta \text{ targ}}(S_{t+1}, a_{t+1}) - \alpha \log \pi(a_{t+1}|s_{t+1})))) \quad (6)$$

actor: one possible version:

$$J_{\pi'}(\phi) = \mathbb{E}_{\tilde{s}_t \sim \mathcal{D}} \left[\mathbb{E}_{\mathbf{a}_t \sim \pi'_\phi} [\alpha \log (\pi'_\phi(\mathbf{a}_t|\hat{s}_t)) - Q_\theta(s_t, \mathbf{a}_t)] \right]$$

$$J_{\pi'}(\phi) = \mathbb{E}_{(\tilde{s}_t, s_t) \sim \mathcal{D}} [\mathbb{E}_{\mathbf{a}_t \sim \pi_\phi} [\frac{\beta}{\text{Var}(Q_\theta(\tilde{s}_t, a))} (\alpha \log (\frac{\beta}{\text{Var}[Q_\theta(\tilde{s}_t, a)]} \pi_\phi(\mathbf{a}_t|\hat{s}_t)/Z(\tilde{s})) - Q_\theta(s_t, \mathbf{a}_t))]]$$